

Muhammad Zaier Ahmad

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SUMMARY

Backend and systems-focused software engineer working in **Go**, **Node.js**, **TypeScript**, and **Lua**. Currently building **fin-erp**, a multi-tenant SaaS ERP covering General Ledger, AP/AR, Inventory, and Procure-to-Pay. Also maintain **penstock**, a zero-dependency workflow orchestration library published on npm, and **capo**, a **Go** compiler for a keyboard-layout DSL with four output targets.

EXPERIENCE

Founder & Lead Engineer, fin-erp (Independent SaaS Product)

Jan. 2026 – Present

- Solo-built a multi-tenant, multi-reseller, multi-currency SaaS ERP end to end: ~**124,000** lines of production JavaScript across **265 REST endpoints**, a **420+ table MS SQL Server** database spanning **9 domain schemas**, and a **117-screen React 19 SPA** written with no data-fetching library and no UI kit
- Engineered the **Node.js / Express 5** backend on a strict **orchestrator** → **pipeline** → **step** → **engine** layer over a **UnitOfWork/BaseRepository** transaction boundary, so every posting (header, lines, GL journal, open item, source update, audit row) commits **atomically or rolls back as one unit**
- Designed **row-level tenant isolation** over a SQL Server hierarchyid organization tree with denormalized ancestor columns, giving **O(1)** tenant → reseller → company → branch resolution behind a 17-middleware context chain and a **deny-wins** permission tree mirrored client-side
- Implemented **Finance** (journal approval state machine with **three posting pipelines**, an **Open Item settlement engine** on persisted computed balances, 12 AP/AR document types, reports sourced strictly from **immutable GL entries**) and **Inventory** (**four costing policies** over a valuation-layer model, landed cost, batch/serial/FEFO expiry, QC holds, reconciliation-gated period close)
- Delivered the full **Procure-to-Pay** chain (requisition → indent → RFQ → weighted **quotation comparison engine** → PO → goods receipt → inspection → return), bridging atomically into AP billing, with a **config-driven tolerance engine** (WARN/BLOCK) behind a privileged override and exception audit
- Hardened the system across **two security and correctness audits**, closing cross-tenant posting holes, session-revocation gaps, concurrency races, and timezone-dependent defects; regression-tested by **40 Vitest integration suites** (~**399 cases**) against a live SQL Server with a **trial-balance invariant** asserted on every posting phase

PROJECTS

penstock, Published npm Library

TypeScript, Vitest, tsup, CI/CD

- Published a **zero-dependency TypeScript library** on npm for composable backend workflows (**use-cases** → **pipelines** → **steps** → **engines**) with **saga-pattern rollback**: automatic reverse-order undo on failure, returning structured step-level results instead of throwing
- Engineered **full generic type inference** with type-level tests, **dual ESM/CJS packaging**, **115 tests** at 95%+ coverage, a CI matrix across **Node 20/22/24**, and supply-chain hardening (**OIDC trusted publishing**, prototype-pollution resistance, zero telemetry)

capo, Keyboard Layout Compiler

Go, Compiler Design, Lexer/Parser, Unicode, XKB

- Designed a declarative **layout DSL** and wrote the **Go** compiler that lowers one . Layout source file to **four output targets**: an **XKB symbols** file, a Linux **vconsole keymap**, a **kanata** behavior skeleton, and an **SVG** cheatsheet, via a hand-rolled **lexer and parser** over a Unicode-aware token model
- Added an **error-collecting validator** that reports every diagnostic in one pass as `file:line:col` with **rune-accurate columns** and **did-you-mean** suggestions, backed by a **normative specification** making diagnostic formats part of the public contract

focus.nvim & scratch.nvim, Neovim Plugins

Lua, libuv (vim.uv), Neovim API, plenary

- focus.nvim**: **non-blocking** Pomodoro tracker on a native **libuv timer handle** (zero polling, no main-thread blocking), with a live **winbar HUD** that saves and restores any pre-existing winbar plus a floating menu with buffer-local keymaps
- scratch.nvim**: per-filetype **singleton scratch buffers** with **stale-handle eviction**, configurable split/float placement, optional `stdpath` persistence, and **tab-completing commands**; both tested headlessly with **plenary busted**

MacroMate, AI Nutrition & Fitness App [FYP]

Flutter, FastAPI, Python, Gemini API, ML

- Shipped a **cross-platform Flutter app** for calorie, macronutrient, and workout tracking on **FastAPI REST services**, with the **Gemini API** for food-image recognition and a pretrained model for meal-plan generation

SKILLS

Languages : Go, JavaScript (ESM), TypeScript, SQL, Python, Lua, C/C++, Dart

Backend & Architecture : Node.js, Express 5, FastAPI, REST API Design, Orchestration & Saga Patterns, Multi-Tenancy, RBAC, JWT Auth, Compiler/Parser Design, Async I/O & Concurrency

Databases : Microsoft SQL Server (hierarchyid, computed columns, filtered indexes, MERGE/HOLDLOCK, rowversion), MySQL, MongoDB

Frontend & Mobile : React 19, Vite, Zustand, React Router, Custom CSS Design Systems, TailwindCSS, Flutter

Testing & Tooling : Vitest, supertest, plenary.nvim, Git, Docker, GitHub Actions, npm Workspaces & Packaging (ESM/CJS), Postman

EDUCATION & CERTIFICATIONS

University of Central Punjab

Mar. 2022 – Mar. 2026

Bachelor of Software Engineering

CGPA: 3.41

Supervised Machine Learning: Regression and Classification

DeepLearning.AI & Stanford Online (Coursera)